
Assignment 3: PCA/FA/Copulas/EVT

QUANTITATIVE FINANCIAL RISK MANAGEMENT

Team 20

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1 Setup and Data

A set of prices for 10 diverse assets were collected, ranging across different types and sectors to make analysis interesting. The data ranges from the start of 2017 to April of 2026. The chosen assets, and some descriptive statistics of their Log>Returns are:-

Table 1: Descriptive Statistics of Daily Log Returns (2017–2026)

Asset	Description	Mean (%)	Std. Dev. (%)	T
<i>Stocks</i>				
ASML	ASML Holding, Technology and Semiconductors	0.1089	2.2167	2319
Shell	Shell, ASE-listed, Energy	0.0364	1.8284	2319
Walmart	Walmart Inc., US retail	0.0815	1.3915	2319
<i>Bonds</i>				
USTB	Vanguard Total Bond Market ETF (BND)	0.0077	0.3589	2319
EUB	iShares Euro Aggregate Bond ETF (SYBB.DE)	−0.0014	0.3486	2319
<i>Indices</i>				
AEX	Amsterdam Exchange Index	0.0310	1.0428	2319
SP500	S&P 500 Index	0.0500	1.1684	2319
<i>Commodities</i>				
Gold	SPDR Gold Shares ETF (GLD)	0.0577	1.0170	2319
Oil	US WTI Crude Oil ETF (BNO)	0.0216	2.5175	2319
Iron	SPDR S&P Metals & Mining ETF (XME)	0.0618	2.0773	2319

Notes: Daily log returns $r_t = \ln(P_t/P_{t-1})$ over January 2017 to April 2026. All series share $T = 2319$ observations after joint cleaning. Data sourced from Yahoo Finance via `yfinance`.

Cleaning: Rows with more than 3 missing assets were dropped. This corresponded to 88 rows, and were observations with either US or Amsterdam exchanged holidays. Post dropping these rows, the only missing values were in the Euro Bond ETF, with a maximum gap size of 1 (i.e. no consecutive missing values). These were forward filled, and then Log>Returns were calculated.

2 Principal Component Analysis

The first step in Principal Component Analysis is the choice between demeaning or standardising data. For the given data, variances vary substantially across assets and classes, and therefore the choice was made to standardise data. As show in Figure 1, simply demeaning would mean high-volatility assets such as ASML and Oil might dominate the principal components. We therefore standardise and look at the correlation matrix.

Log Return Covariance and Correlation | 2017-2026

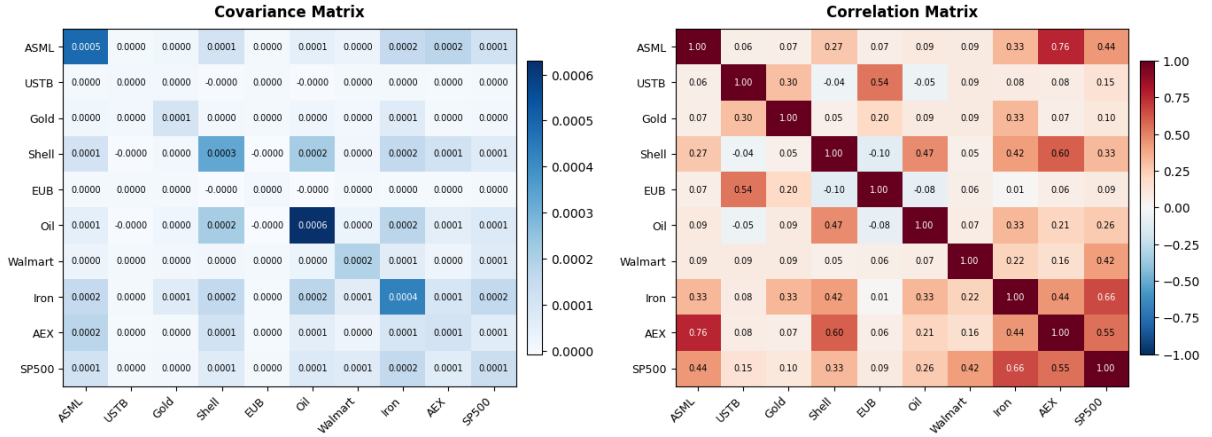


Figure 1: Covariance and Variance Heatmap of daily log returns - Motivating Demeaning vs. Standardising

To select the number of principal components, we look at Figure 2. Kaiser's rule indicates a cutoff at 4 principal components, but the fifth component lies right below the threshold and the cumulative variance curve suggests a gradual drop off. With the low number of assets, Kaiser's rule is also restrictive, and hence we extend to 5 principal components for analysis, which explain 81% of total variance.

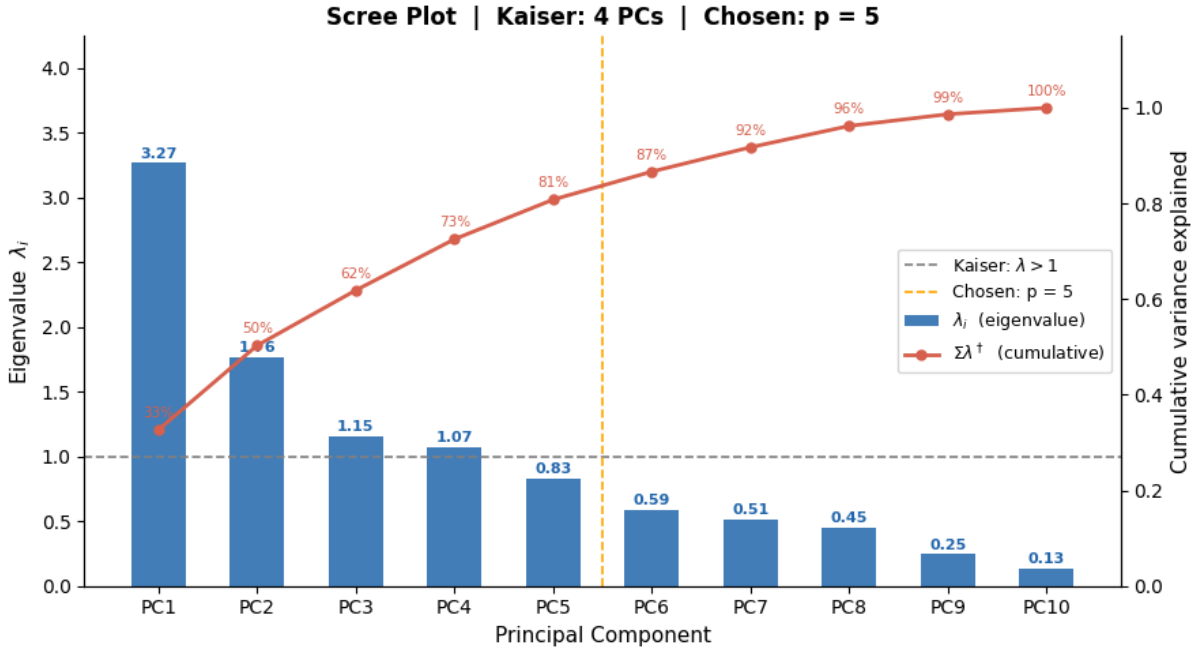


Figure 2: Scree Plot - Factor Selection

Figure 3 decomposes variance explained per asset by each principal component, and the R^2 per asset. PC1 loads mainly on equities and indices - ASML, AEX, SP500 - and perhaps uninventively on Iron. PC2 appears to load mainly on the bonds. PC3 remains hard to interpret, perhaps an EU tilt. PC4 isolates Walmart, with previous factors explaining a low amount of it's variance due to it's US focus and unique sector. PC5 once appears to load on commodities. For

PC2 and PC5 we see that bonds and commodities have some cross-movement due to their role as equity hedges. The loadings are generally interpretable and appear to make sense.

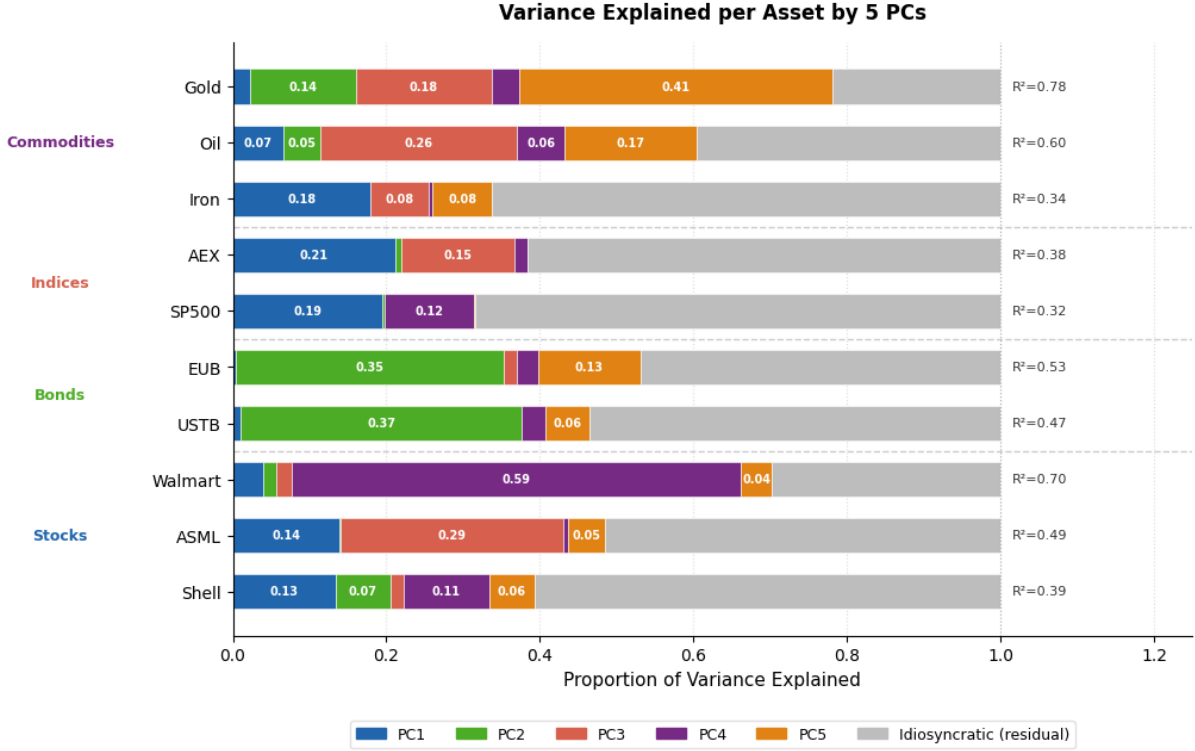


Figure 3: Variance Explained per Asset - Decomposing the Factors

The R^2 remains low, mainly for the indices and Iron. This makes sense, as the indices include a broad market exposure across sectors which is hard to capture with few principal components, and unlike other commodities, Iron does not serve as an immediate hedge. For the low number of assets, using PCA does not particularly make sense, as there is no need for dimension-reduction.

3 Factor Analysis

To perform factor analysis without PCA, we use fundamental factor-mimicking portfolios instead of statistically constructed principal components. Since the asset universe contains equities, bonds and commodities, we define three interpretable factors:

$$F_t^{Eq}, \quad F_t^{Bond}, \quad F_t^{Com}.$$

The equity factor is the average return of ASML, Shell, Walmart, AEX and S&P 500. The bond factor is the average return of USTB and EUB. The commodity factor is the average return of Gold, Oil and Iron. For each asset, the asset itself is excluded from its own group factor to avoid mechanically explaining the asset by itself.

Each standardized asset return is regressed on the three standardized factors:

$$\tilde{r}_{i,t} = \alpha_i + \beta_{i,Eq} F_t^{Eq} + \beta_{i,Bond} F_t^{Bond} + \beta_{i,Com} F_t^{Com} + \epsilon_{i,t}.$$

This gives a non-PCA factor model with three economically meaningful factors.

Table 2: Fundamental factor loadings

Asset	Equity	Bond	Commodity
ASML	0.532	0.049	-0.061
Shell	0.229	-0.133	0.427
Walmart	0.140	0.073	0.103
USTB	0.023	0.534	0.070
EUB	0.025	0.538	-0.052
AEX	0.868	0.026	-0.089
SP500	0.455	0.098	0.301
Gold	-0.069	0.300	0.286
Oil	0.228	-0.123	0.206
Iron	0.487	0.010	0.259

Table 2 and Figure 4 show that the estimated loadings are economically intuitive. AEX, ASML and S&P 500 have the strongest positive exposure to the equity factor, while USTB and EUB mainly load positively on the bond factor. Shell loads positively on the commodity factor, reflecting its link to energy markets. The negative loadings are also meaningful. For example, Gold has a negative loading on the equity factor, which is consistent with its role as a safe-haven asset that can move differently from equity markets. Shell and Oil have negative loadings on the bond factor, suggesting that they are less aligned with defensive fixed-income movements. Overall, both the positive and negative signs support an economically sensible interpretation of the factor model.

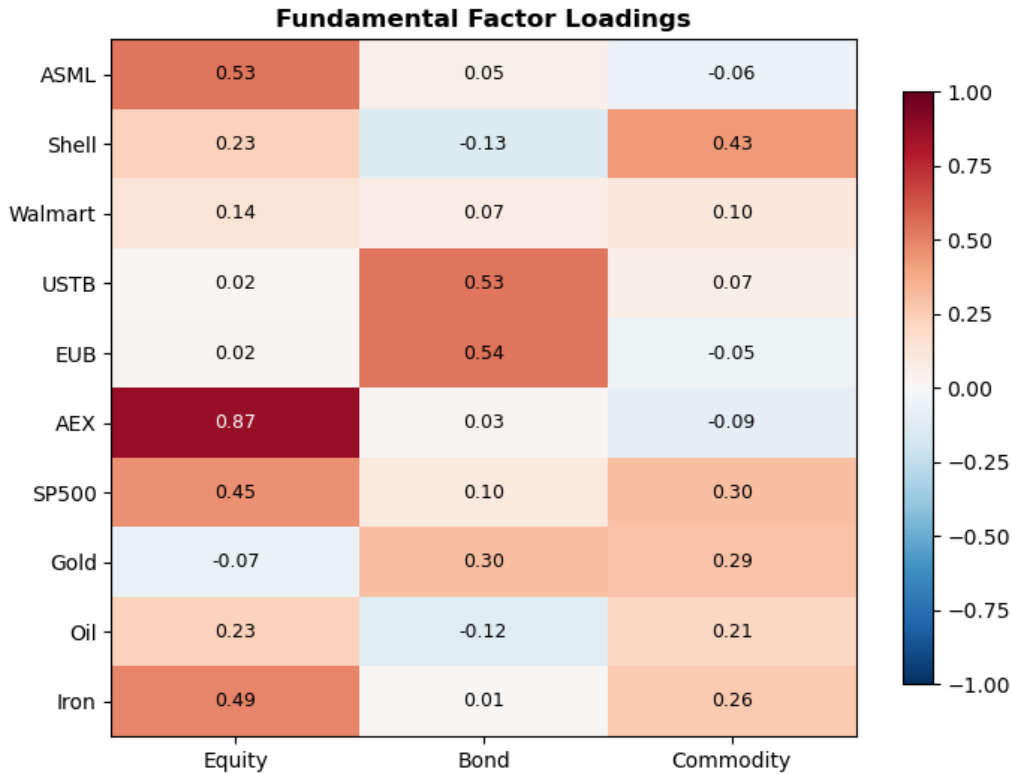


Figure 4: Loadings of each asset on the fundamental factors

The explanatory power differs strongly across assets as per Table 3. The model explains AEX best, with an R^2 of 0.684. It also explains SP500, Iron and Shell reasonably well. Walmart, Oil and Gold are less well explained by these broad factors, suggesting larger idiosyncratic components.

Table 3: Explained variance of the fundamental factor model

Asset	R^2	Idiosyncratic share
AEX	0.684	0.316
SP500	0.442	0.558
Iron	0.388	0.612
Shell	0.320	0.680
USTB	0.294	0.706
EUB	0.288	0.712
ASML	0.255	0.745
Gold	0.149	0.851
Oil	0.144	0.856
Walmart	0.052	0.948

Figure 5 compares the explanatory power of the fundamental factor model with the PCA model. PCA explains more variation for most assets because its factors are constructed to maximize variance. The average asset-level R^2 is 0.302 for the fundamental factor model, compared with 0.500 for the PCA model. However, the fundamental factor model is easier to interpret because the factors have clear economic meanings.

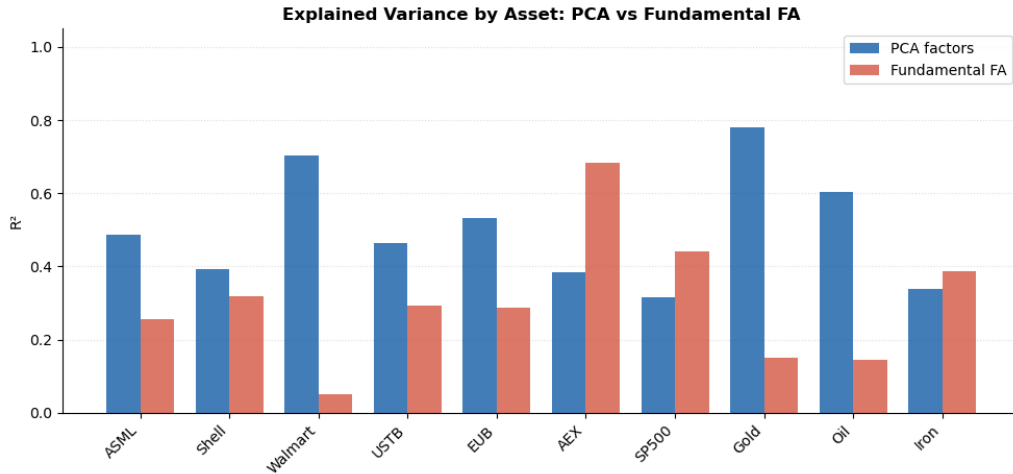


Figure 5: Asset-level R^2 : PCA factors versus fundamental factors

Figure 6 shows the residual correlations after removing the three fundamental factor effects. The average absolute correlation decreases from 0.224 before the factor model to 0.133 after the factor model. This means that the factors capture part of the common dependence structure, although some residual co-movement remains.

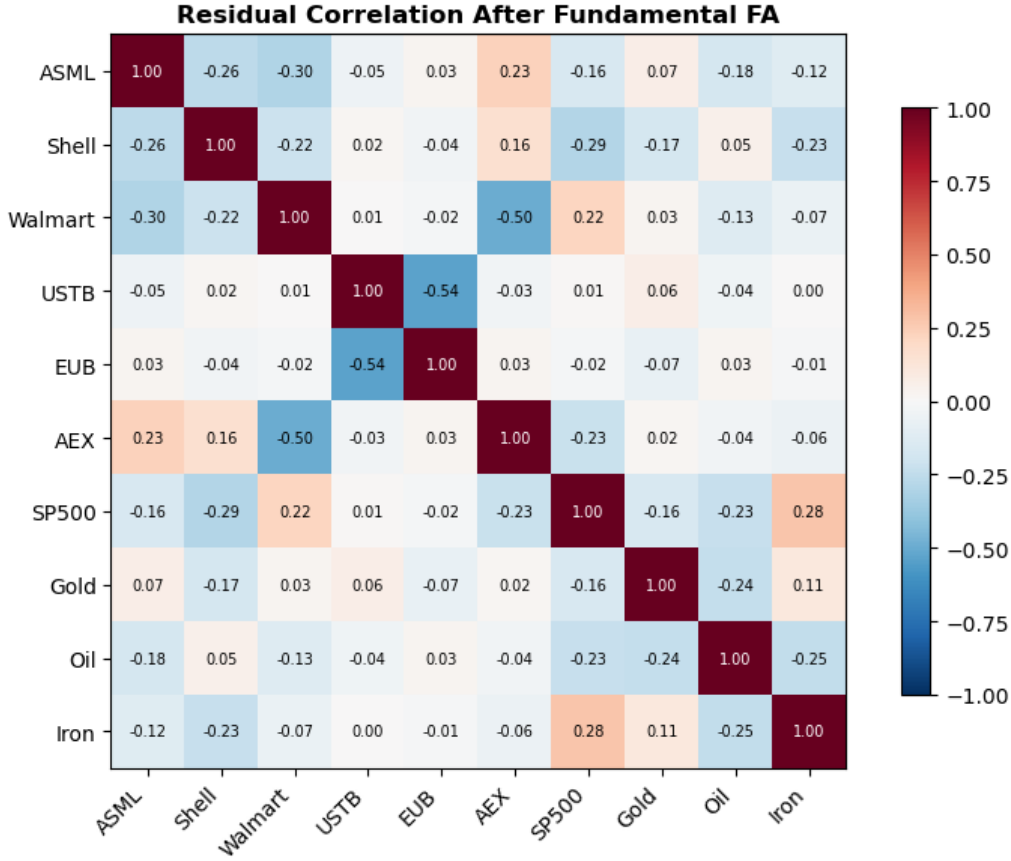


Figure 6: Residual correlations after removing the fundamental factor effects

Overall, the PCA and fundamental factor approaches are useful for different purposes. PCA is better if the objective is to explain as much return variation as possible, since the factors are chosen statistically to maximize variance. This is why the PCA model achieves the higher average R^2 . However, the PCA factors are harder to interpret economically. The fundamental factor model explains less variation, but it gives a clearer economic story: asset returns are mainly driven by exposure to equity, bond and commodity risk. Therefore, PCA is stronger from a purely statistical perspective, while the fundamental factor approach is more useful for interpretation and reporting.

4 Copulas

We analyse three bivariate return pairs: SP500-ASML (high-risk equity vs. high-risk tech), Shell-EUB (energy equity vs. sovereign bond), and Oil-USTB (commodity vs. US treasury bond). The latter two mix high-risk with low-risk assets across different asset classes. Marginals were selected based on univariate QQ-plot diagnostics (Figure 7): bond returns (EUB, USTB) were modelled with Gaussian marginals, while equity and commodity returns were modelled with Student- t marginals, consistent with their heavier tails.

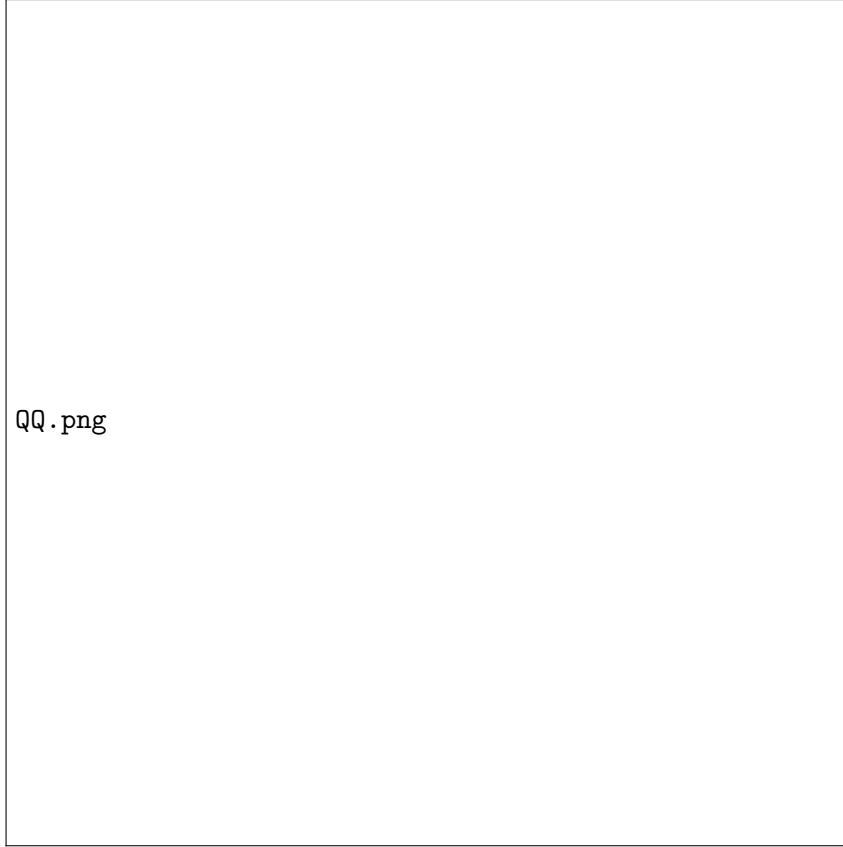


Figure 7: Normal and Student- t QQ plots for selected assets. Bond returns align well with the normal distribution; equity and commodity returns exhibit heavier tails, motivating the Student- t marginal.

For each pair, observed returns were transformed into pseudo-observations $u_i = F_X(x_i)$, $v_i = F_Y(y_i)$, and a Gaussian copula and a Student- t copula were fitted separately. The Gaussian copula provides a zero-tail-dependence benchmark; the Student- t copula allows symmetric tail dependence through an additional degrees-of-freedom parameter ν_c . Parameter precision was assessed via $B = 250$ bootstrap replications.

4.1 Marginal and copula parameter estimates

Table 4 reports fitted marginal and copula parameters. For **SP500-ASML**, both marginals are heavy-tailed ($\hat{\nu} = 2.61$ and 3.74) and copula dependence is positive ($\hat{\rho} \approx 0.50$). For **Shell-EUB** and **Oil-USTB** dependence is weakly negative ($\hat{\rho} \approx -0.14$ for both). The fitted Student- t copula degrees of freedom are 4.1, 10.6, and 8.6 respectively, indicating the heaviest copula tails for **SP500-ASML**.

Table 4: Marginal and copula parameter estimates.

Pair	Model	Parameter	Estimate
SP500-ASML	Marginal SP500	ν	2.6092
	Marginal SP500	μ	0.0010
	Marginal SP500	σ	0.0065
	Marginal ASML	ν	3.7409
	Marginal ASML	μ	0.0014
	Marginal ASML	σ	0.0156
	Gaussian copula	ρ	0.4995
	Student- t copula	ρ	0.4995
	Student- t copula	ν_c	4.1000
Shell-EUB	Marginal Shell	ν	2.6517
	Marginal Shell	μ	0.0010
	Marginal Shell	σ	0.0101
	Marginal EUB	μ	−0.0000
	Marginal EUB	σ	0.0035
	Gaussian copula	ρ	−0.1389
	Student- t copula	ρ	−0.1389
	Student- t copula	ν_c	10.6000
Oil-USTB	Marginal Oil	ν	3.4257
	Marginal Oil	μ	0.0015
	Marginal Oil	σ	0.0158
	Marginal USTB	μ	0.0001
	Marginal USTB	σ	0.0036
	Gaussian copula	ρ	−0.1423
	Student- t copula	ρ	−0.1423
	Student- t copula	ν_c	8.6000

4.2 Model selection: AIC and BIC

Table 5 reports the copula log-likelihood, AIC, and BIC for each pair.

Table 5: Copula log-likelihood, AIC, and BIC by pair and model. Lower AIC/BIC indicates a better-fitting model after penalising for complexity.

Pair	Model	k	Log-lik	AIC	BIC
SP500-ASML	Gaussian	1	296.22	−590.44	−584.69
	Student- t	2	363.35	−722.70	−711.20
Shell-EUB	Gaussian	1	18.44	−34.88	−29.13
	Student- t	2	43.85	−83.71	−72.21
Oil-USTB	Gaussian	1	10.96	−19.91	−14.17
	Student- t	2	43.97	−83.94	−72.44

Across all three pairs the Student- t copula is preferred by both AIC and BIC, despite the additional parameter penalty. The improvement is largest for **SP500-ASML**, where the log-likelihood increases from 296.2 to 363.4 — a gain of 67 log-likelihood units for one extra parameter — which strongly supports a heavy-tailed copula for this pair. For **Shell-EUB** and **Oil-USTB** the improvement is more modest (log-likelihood gains of 25 and 33 respectively), consistent with the large estimated ν_c in those pairs: the Student- t copula is still selected, but it is close to Gaussian in tail behaviour.

4.3 Dependence measures and tail dependence

Table 6 compares empirical dependence measures with those implied by the fitted copulas. SP500-ASML exhibits the strongest positive dependence (empirical Pearson 0.44, Kendall’s τ 0.33, Spearman’s ρ_s 0.47). Both copula models reproduce rank dependence well. Critically, only the Student- t copula captures substantial tail dependence ($\hat{\lambda} \approx 0.248$), compared to the empirical joint-tail rate of 0.31 (lower) and 0.28 (upper). The Gaussian copula assigns zero tail dependence by construction, regardless of ρ .

For Shell-EUB and Oil-USTB dependence is weakly negative. The Student- t copula permits non-zero tail dependence but the estimates remain negligible (0.002 and 0.005), largely because the fitted ν_c is large. A notable feature of Oil-USTB is that Spearman’s ρ_s (-0.132) diverges from Kendall’s τ (-0.091) across all three estimation sources. This reflects concentration of negative co-movement in the middle of the return distribution rather than at extremes, consistent with an inflation-channel relationship that weakens during macro stress.

Table 6: Empirical and model-implied dependence measures.

Pair	Source	Pearson	Kendall’s τ	Spearman’s ρ_s	$\hat{\lambda}_L$	$\hat{\lambda}_U$
SP500-ASML	Empirical	0.4359	0.3330	0.4686	0.3105	0.2846
SP500-ASML	Gaussian	0.4485	0.3379	0.4889	0.0000	0.0000
SP500-ASML	Student- t	0.4931	0.3361	0.4723	0.2478	0.2478
Shell-EUB	Empirical	-0.1034	-0.0887	-0.1301	0.0604	0.0517
Shell-EUB	Gaussian	-0.1295	-0.0940	-0.1407	0.0000	0.0000
Shell-EUB	Student- t	-0.1373	-0.0871	-0.1289	0.0022	0.0022
Oil-USTB	Empirical	-0.0492	-0.0909	-0.1322	0.0690	0.0862
Oil-USTB	Gaussian	-0.1412	-0.0962	-0.1439	0.0000	0.0000
Oil-USTB	Student- t	-0.1428	-0.0900	-0.1328	0.0054	0.0054

Empirical $\hat{\lambda}$ is computed as the joint exceedance rate at the 5% threshold. Copula rows report the analytical tail dependence coefficient.

4.4 Precision of the estimates

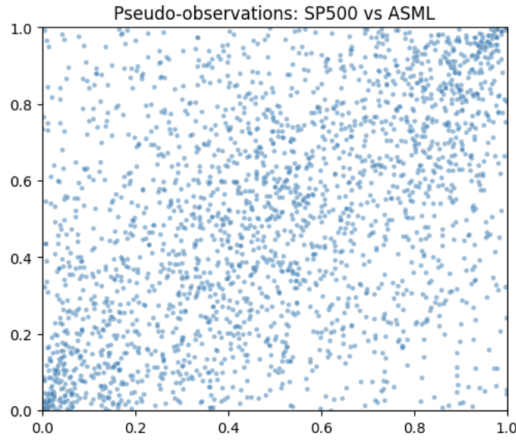
Table 7 reports bootstrap standard errors and 95% confidence intervals for the copula parameters. The dependence parameter ρ is estimated precisely in all pairs: for SP500-ASML the 95% CI is [0.468, 0.531], firmly excluding zero; for Shell-EUB and Oil-USTB the intervals are entirely negative. The Student- t copula degrees-of-freedom ν_c carries more uncertainty, particularly for Shell-EUB (CI: [8.1, 16.1]), but the ordering across pairs is stable and informative.

Table 7: Bootstrap precision for copula parameters ($B = 250$).

Pair	Parameter	Estimate	Std. error	2.5% CI	97.5% CI
SP500-ASML	Gaussian ρ	0.5000	0.0167	0.4680	0.5305
SP500-ASML	Student- t ρ	0.5000	0.0167	0.4680	0.5305
SP500-ASML	Student- t ν_c	4.3480	0.5401	3.6000	5.6000
Shell-EUB	Gaussian ρ	-0.1380	0.0228	-0.1819	-0.0893
Shell-EUB	Student- t ρ	-0.1380	0.0228	-0.1819	-0.0893
Shell-EUB	Student- t ν_c	11.2400	2.0384	8.1000	16.1000
Oil-USTB	Gaussian ρ	-0.1416	0.0255	-0.1902	-0.0929
Oil-USTB	Student- t ρ	-0.1416	0.0255	-0.1902	-0.0929
Oil-USTB	Student- t ν_c	8.7020	1.5557	6.6000	12.3750

4.5 Visual representation

Figures 8-16 present the pseudo-observation scatterplots alongside contour and 3D surface plots of the fitted Student- t copula density. SP500-ASML shows clear positive dependence with visible corner concentration under the Student- t copula, indicating non-negligible tail dependence. In contrast, Shell-EUB and Oil-USTB exhibit weak negative dependence and comparatively diffuse density surfaces, consistent with near-zero tail dependence and more Gaussian-like behaviour.

**Figure 8:** Pseudo-observations for SP500-ASML.

StudentT copula density surface: SP500-ASML

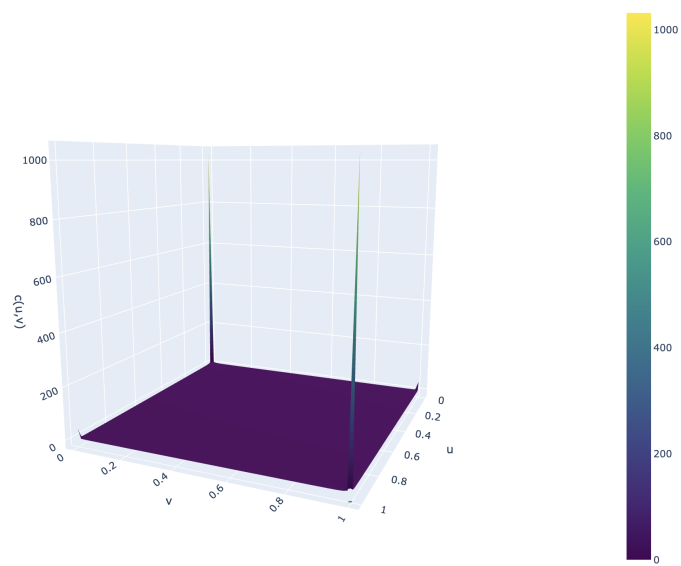


Figure 9: Student- t density for SP500-ASML.

Gaussian copula density surface: SP500-ASML

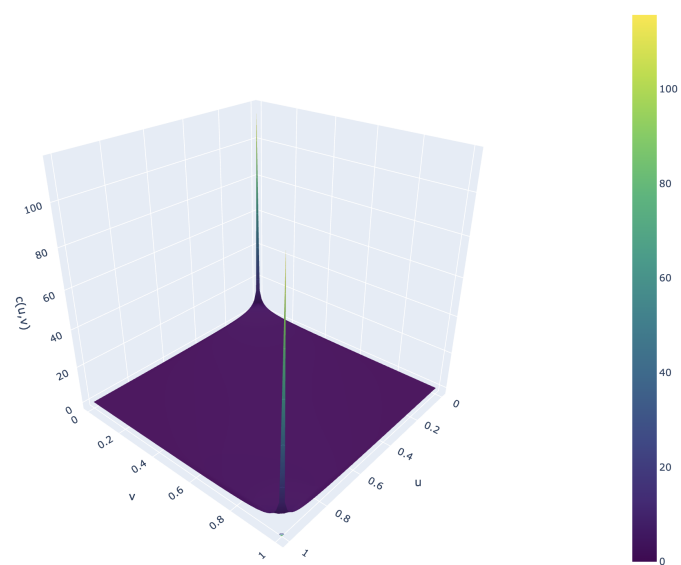


Figure 10: Gaussian density for SP500-ASML.

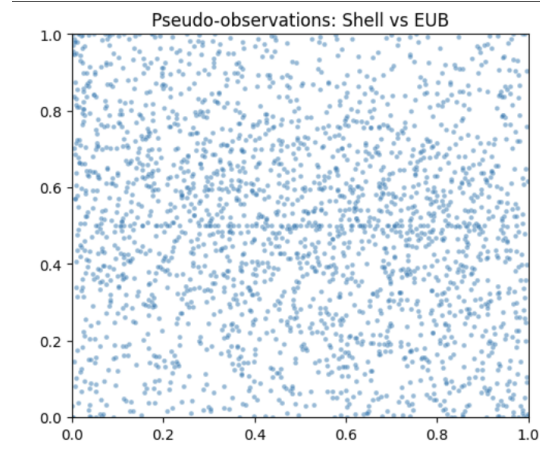


Figure 11: Pseudo-observations for Shell-EUB.

StudentT copula density surface: Shell-EUB

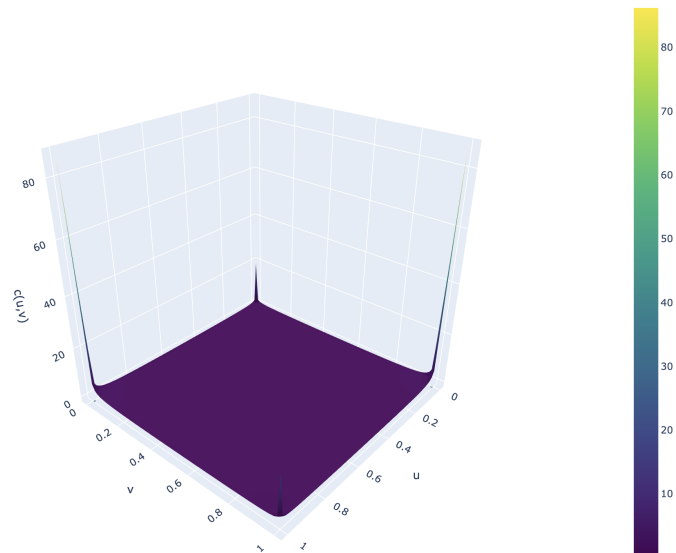


Figure 12: Student- t density for Shell-EUB.

Gaussian copula density surface: Shell-EUB

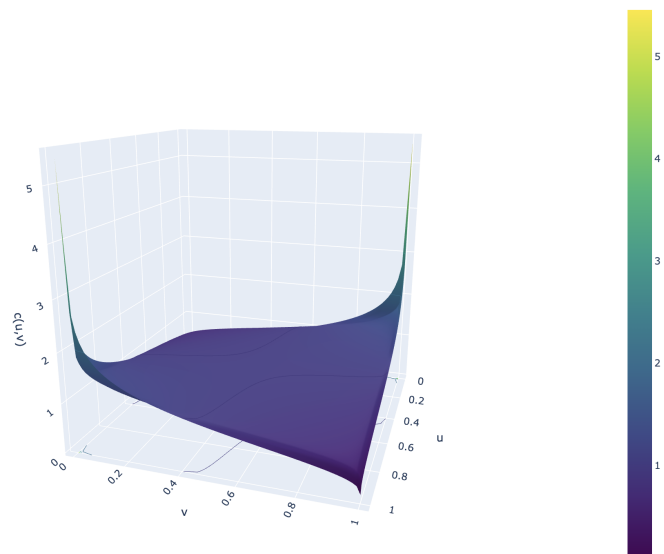


Figure 13: Gaussian density for Shell-EUB.

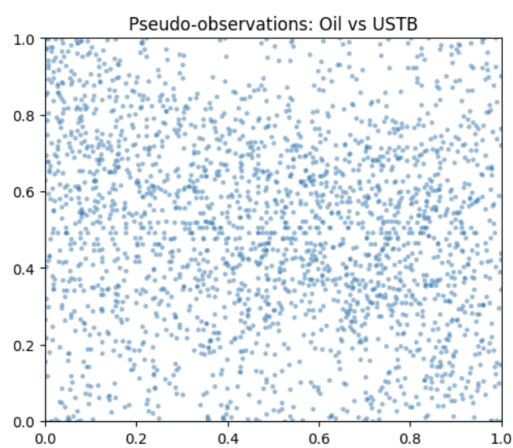


Figure 14: Pseudo-observations for Oil-USTB.

StudentT copula density surface: Oil-USTB

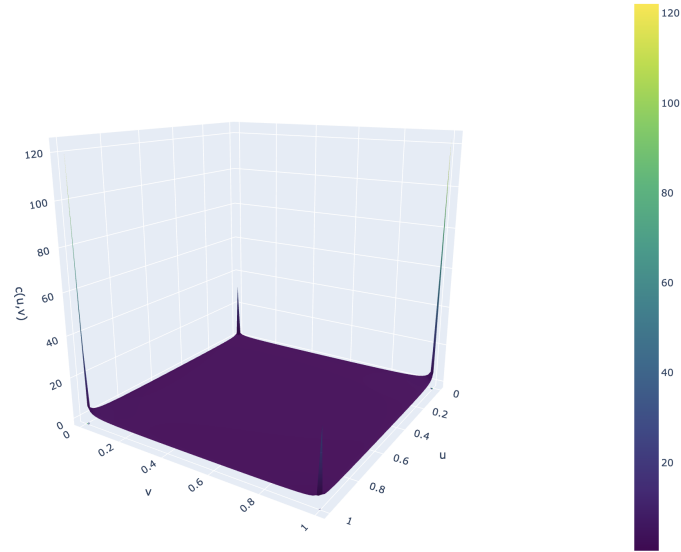


Figure 15: Student- t density for Oil-USTB.

Gaussian copula density surface: Oil-USTB

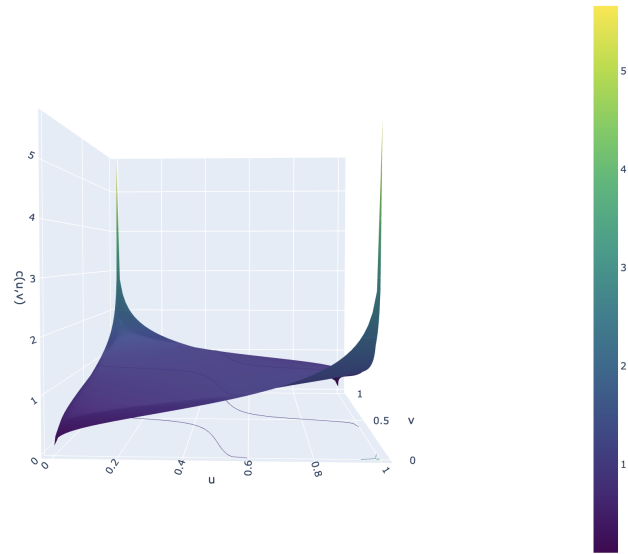


Figure 16: Gaussian density for Oil-USTB.

5 Extreme Value Theory

The final part of the analysis studies extreme losses using Extreme Value Theory (EVT). The assignment asks to identify the asset with the heaviest tails and then apply a Peak-over-Threshold approach to that asset. Since risk management is mainly concerned with large negative returns, the analysis is carried out on losses, defined as

$$L_t = -r_t.$$

A high value of L_t therefore corresponds to a large negative daily return.

5.1 Tail diagnostics and asset selection

As a first diagnostic, Student- t QQ plots were made for all ten assets. The purpose is to check whether the empirical tails are heavier than what is captured by a fitted Student- t distribution. The QQ plots show that most equity and commodity assets have visibly heavy tails, especially Oil, Shell and the S&P 500. However, the strongest excess kurtosis is observed for USTB.

Student-t QQ plots of daily log returns

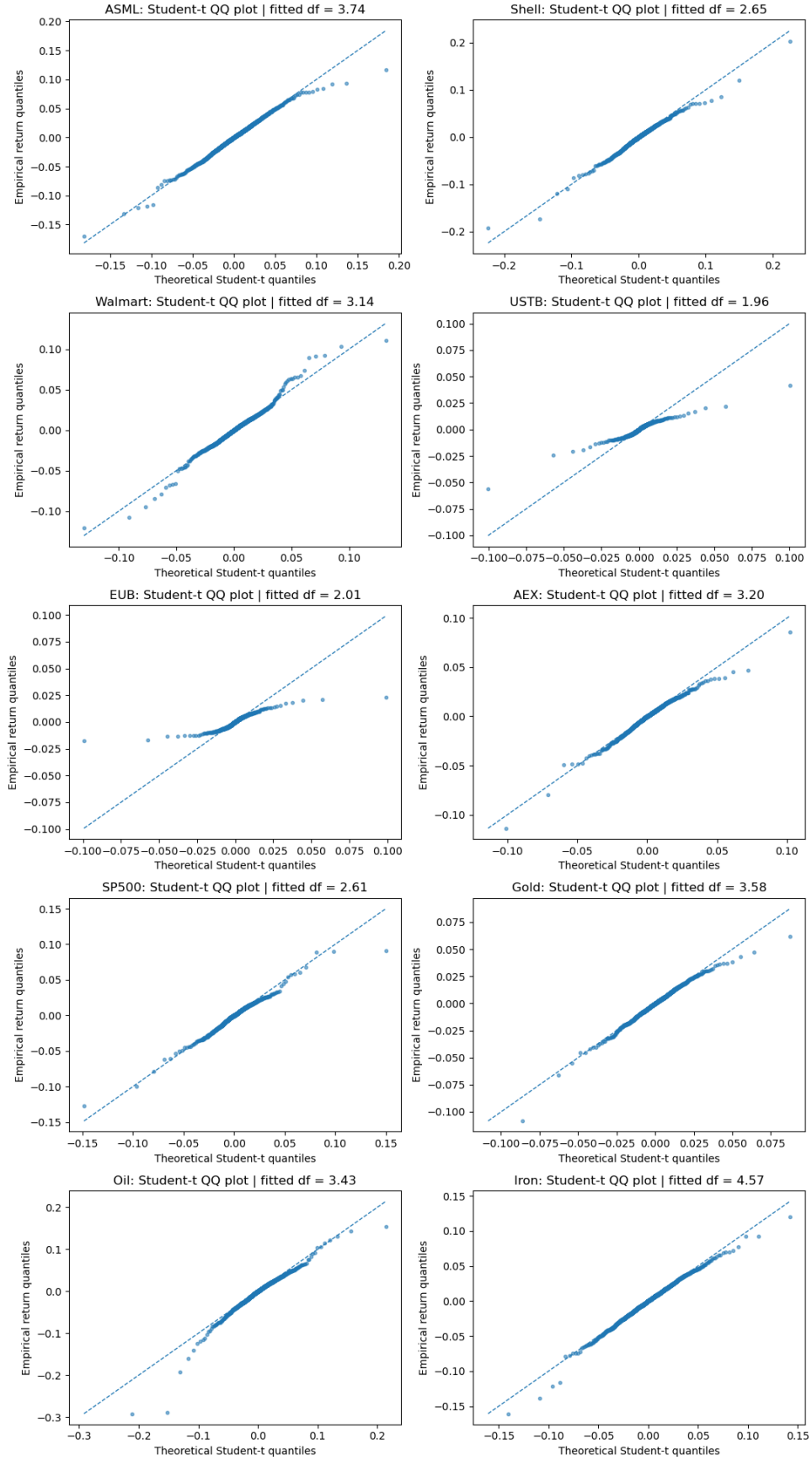


Figure 17: Student- t QQ plots of daily log returns for all assets.

Table 8 reports the main tail diagnostics. USTB has the highest excess kurtosis, equal to 36.883. This is much larger than the value for Oil, which has the second highest excess kurtosis of 20.678. Therefore, USTB is selected for the EVT analysis. This choice is not based on volatility, but on tail thickness. Although USTB has low daily volatility, its return distribution contains some unusually large negative observations relative to its usual movement.

Table 8: Tail diagnostic summary of daily log returns

Asset	Mean	Std. Dev.	Skewness	Excess Kurtosis	Loss 95%	Loss 99%
USTB	0.000077	0.003572	-1.131628	36.883156	0.005129	0.008884
Oil	0.000215	0.025094	-1.733974	20.678446	0.035880	0.073721
Shell	0.000362	0.018181	-0.567059	19.448471	0.027199	0.052325
SP500	0.000496	0.011698	-0.659160	16.463213	0.017217	0.034235
Walmart	0.000809	0.013897	0.041392	13.576703	0.018997	0.032835
AEX	0.000308	0.010383	-0.872542	11.654173	0.016203	0.030748
Gold	0.000573	0.010117	-0.770276	9.124447	0.015458	0.029690
Iron	0.000614	0.020728	-0.467060	4.673549	0.031838	0.055884
ASML	0.001081	0.022067	-0.387640	4.559680	0.034609	0.059871
EUB	-0.000014	0.003486	0.321223	4.156211	0.005600	0.009405

5.2 Peak-over-Threshold model

For USTB, the Peak-over-Threshold method is applied to losses above a high threshold u . The exceedances are defined as

$$Y_t = L_t - u \quad \text{given that} \quad L_t > u.$$

For sufficiently high thresholds, EVT suggests that the exceedances can be approximated by a Generalized Pareto Distribution (GPD):

$$G(y; \xi, \beta) = 1 - \left(1 + \frac{\xi y}{\beta}\right)^{-1/\xi}, \quad y > 0,$$

where ξ is the shape parameter and β is the scale parameter. The shape parameter is especially important because it determines the heaviness of the tail. A positive ξ implies a heavy tail.

The threshold cannot be chosen mechanically. A lower threshold gives more exceedances, but may include observations that are not truly extreme. A higher threshold focuses more directly on the tail, but leaves fewer observations for estimation. To assess this trade-off, we inspected the mean excess plot and the stability of the estimated GPD parameters across several thresholds.

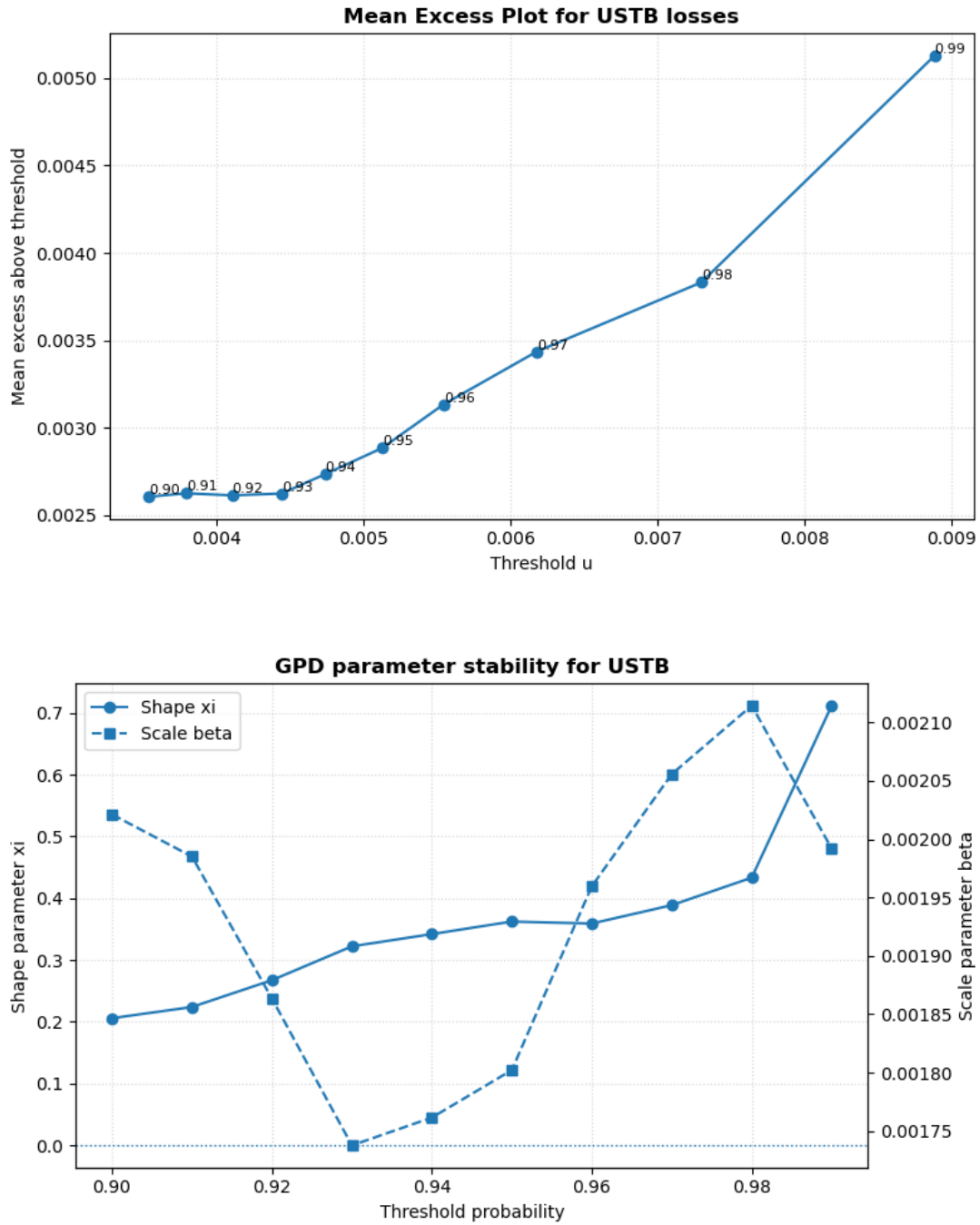


Figure 18: Threshold diagnostics for USTB losses: mean excess plot and GPD parameter stability.

The mean excess plot increases roughly with the threshold, suggesting heavy-tailed behaviour. The estimated shape parameter is also reasonably stable around the 95% threshold, before becoming more unstable at very high thresholds. At the 99% threshold, only 24 exceedances remain, which is too few for a reliable fit. Therefore, the 95% loss quantile is used as the threshold. This gives 116 exceedances, corresponding to approximately 5% of the sample.

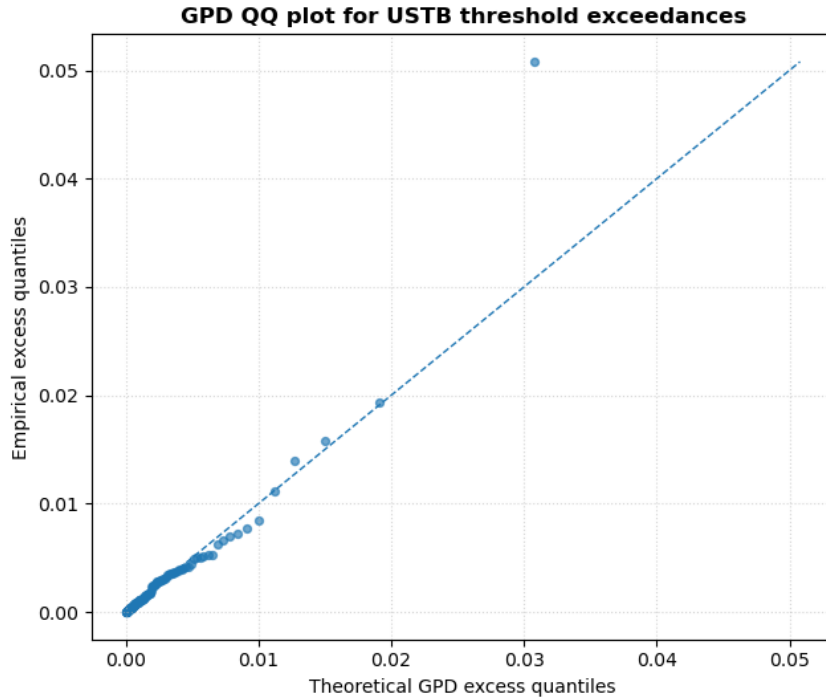
Table 9: Threshold diagnostics for USTB losses

Threshold Prob.	Threshold u	Exceedances	Exceedance Share	ξ	β	Mean Excess
0.90	0.003544	232	0.100043	0.205856	0.002021	0.002605
0.91	0.003796	209	0.090125	0.224058	0.001985	0.002625
0.92	0.004110	186	0.080207	0.267444	0.001863	0.002613
0.93	0.004446	163	0.070289	0.322248	0.001738	0.002623
0.94	0.004742	140	0.060371	0.342072	0.001761	0.002734
0.95	0.005129	116	0.050022	0.362374	0.001802	0.002884
0.96	0.005546	93	0.040103	0.359153	0.001960	0.003133
0.97	0.006180	70	0.030185	0.388843	0.002056	0.003436
0.98	0.007298	47	0.020267	0.433430	0.002114	0.003832
0.99	0.008884	24	0.010349	0.711940	0.001992	0.005130

Using the 95% threshold, the fitted GPD parameters are

$$\hat{\xi} = 0.362374, \quad \hat{\beta} = 0.001802.$$

The positive value of $\hat{\xi}$ indicates a heavy-tailed loss distribution. This means that extreme losses decline relatively slowly in the tail, so risk measures such as VaR and ES can increase quickly at high confidence levels.

**Figure 19:** GPD QQ plot for USTB threshold exceedances.

The GPD QQ plot shows that the fitted distribution describes most exceedances reasonably well. The largest observation is still difficult to fit exactly, but this is common in EVT because the method is specifically focused on rare tail observations. Overall, the GPD fit appears acceptable for estimating high-level tail risk.

5.3 VaR and Expected Shortfall

The fitted GPD is then used to estimate Value-at-Risk and Expected Shortfall. For a confidence level α , the POT-GPD VaR estimate is

$$VaR_\alpha = u + \frac{\hat{\beta}}{\hat{\xi}} \left[\left(\frac{N_u/N}{1-\alpha} \right)^{\hat{\xi}} - 1 \right],$$

where N is the total number of observations and N_u is the number of threshold exceedances. The corresponding Expected Shortfall is

$$ES_\alpha = \frac{VaR_\alpha + \hat{\beta} - \hat{\xi}u}{1 - \hat{\xi}},$$

which is finite because $\hat{\xi} < 1$.

Table 10 compares the EVT estimates with historical simulation and a GARCH(1,1) model with Student- t innovations. The comparison is useful, but the estimates are not perfectly the same object. Historical simulation and EVT are unconditional full-sample tail estimates, while the GARCH model gives a conditional one-step-ahead estimate based on the most recent volatility forecast.

Table 10: VaR and ES comparison for USTB losses

Confidence Level	Method	VaR	ES
95.0%	Historical simulation	0.005129	0.008012
95.0%	EVT POT-GPD	0.005130	0.007956
95.0%	GARCH + Student- t	0.004157	0.005631
97.5%	Historical simulation	0.006537	0.010303
97.5%	EVT POT-GPD	0.006550	0.010184
97.5%	GARCH + Student- t	0.005172	0.006654
99.0%	Historical simulation	0.008884	0.014013
99.0%	EVT POT-GPD	0.009068	0.014133
99.0%	GARCH + Student- t	0.006500	0.008031

The EVT estimates are very close to historical simulation at the 95% and 97.5% levels. At the 99% level, EVT gives a slightly higher VaR and ES than historical simulation. This is expected because EVT extrapolates the tail beyond the observed quantiles using the fitted GPD, while historical simulation is fully determined by the realised sample. The GARCH + Student- t estimates are lower across all confidence levels. This is likely because the GARCH estimate is conditional on the latest volatility state, while the historical and EVT estimates reflect the full sample period, including more stressed observations.

Overall, the EVT results suggest that USTB has low normal-day volatility but a very heavy left tail. This is an important distinction: the asset may appear safe based on standard deviation, but its extreme negative return behaviour is less benign. For USTB, the fitted GPD shape parameter is positive and the 99% EVT Expected Shortfall is around 1.41% for one trading day. Therefore, EVT adds value by focusing directly on the loss tail rather than on average volatility.

Appendix

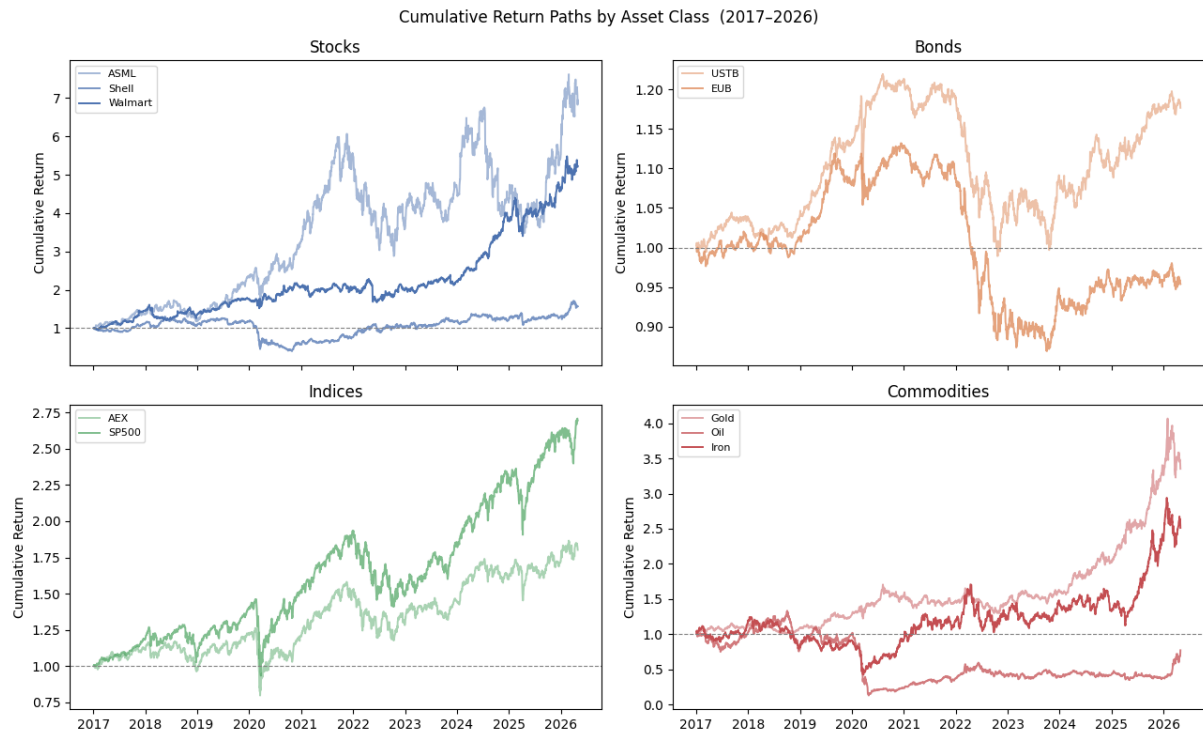


Figure 20: Cumulative Returns of Assets